

UNIT-2: Magneto-statics

- Introduction to magnetic field vectors, Bio-Savart and Ampere's law, Magnetic scalar and vector potential
- Self and Mutual Inductance energy stored in magnetic field, Boundary condition.
- Analogy between electric and magnetic field, magneto motive force, reluctance.

Electric field vector $\mathbf{E}$ is the basic quantity in electrostatics, the magnetic-flux density or magnetic induction $\mathbf{B}$ plays a fundamental role in magneto statics. Another fundamental quantity is the magnetic dipole $\boldsymbol{\mu}$

$$\mathbf{N} = \boldsymbol{\mu} \times \mathbf{B}.$$

It was also established that there is a connection between electrical currents and magnetic fields. As will soon be shown, a current density $\mathbf{J}$ (or simply a current I) is a source of magnetic-flux density. Since the current density is defined as the amount of charge that flows through a cross-section per unit of time (units of Coulombs per square meter-second), the conservation of charge requires that the so-called *continuity equation* be satisfied

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0}$$

Because magnetostatics is concerned with steady-state currents, limit to the following equation

$$\nabla \cdot \mathbf{J} = 0$$

Stationary charges produce electric fields that are constant in time. *Steady currents* produce magnetic fields that are also constant in time.

Stationary charges $\Rightarrow$ constant electric fields; electrostatics.
Steady currents $\Rightarrow$ constant magnetic fields; magnetostatics.

Steady current means that a continuous flow that goes on forever without change and without charge piling up anywhere. They represent *suitable approximations* as long as the fluctuations are reasonably slow.

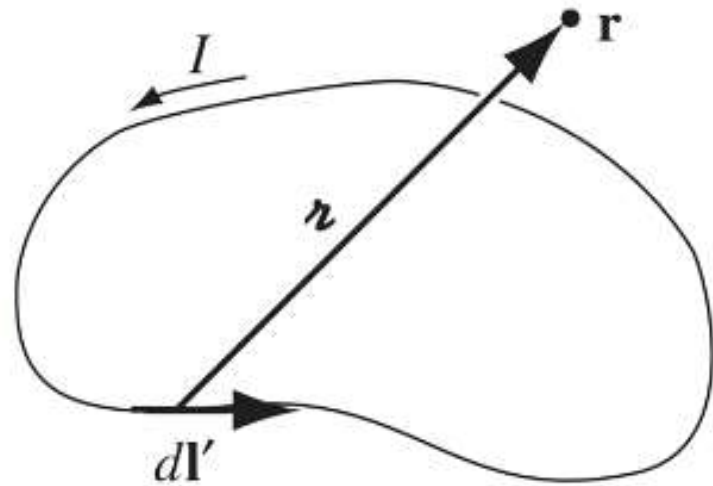
$$\nabla \cdot \mathbf{J} = 0$$

Biot-Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl' = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

The integration is along the current path, in the direction of the flow.

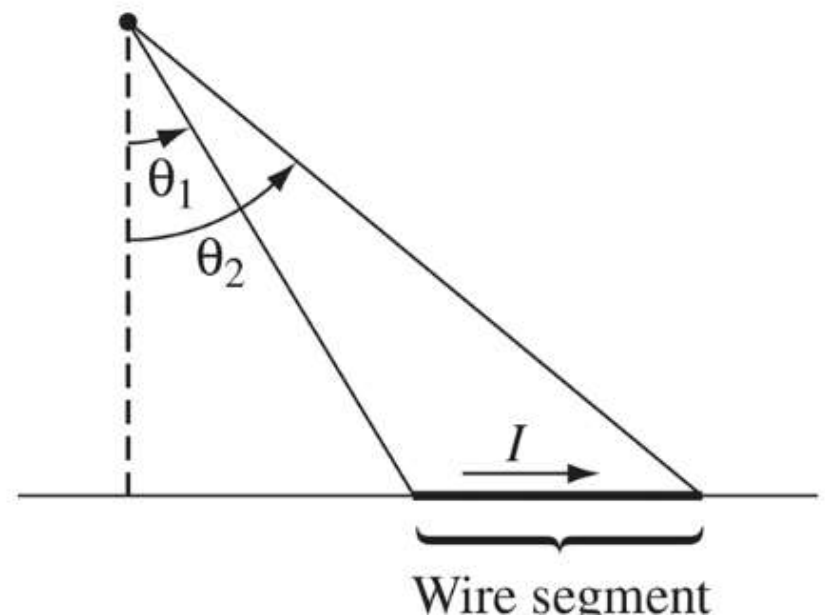
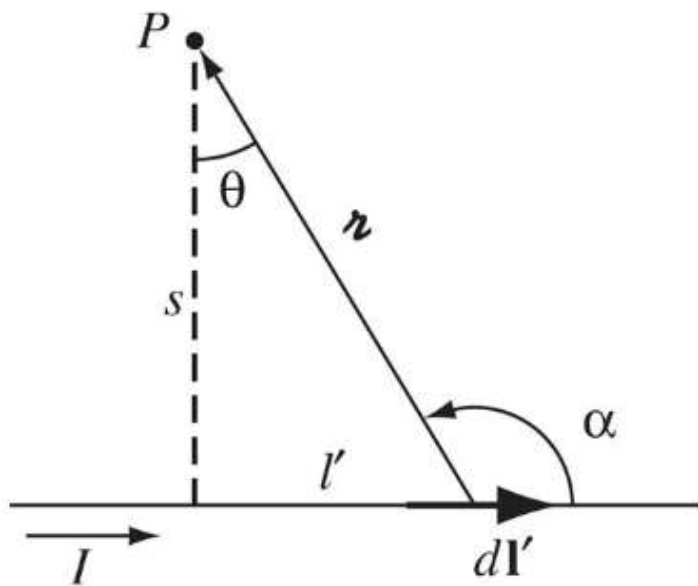
μ_0 : the *permeability of free space*.



Definition of magnetic field $\mathbf{B}$: newtons per ampere-meter or **tesla** (T). $1 \text{ T} = 1 \text{ N}/(\text{A}\cdot\text{m})$

The Biot-Savart law plays a role analogous to Coulomb's law in electrostatics.

Q. Find the magnetic field a distance s from a long straight wire carrying a steady current I



Sol :

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl' = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

Then, determine the suitable coordinate: cylindrical coordinate (s, ϕ, z) .

In the diagram, $(d\mathbf{l}' \times \hat{\mathbf{r}})$ points out of page $\hat{\phi}$ and has the magnitude $dl' \sin \alpha = dl' \cos \theta$

$$l' = s \tan \theta \Rightarrow dl' = s \sec^2 \theta d\theta \quad \text{and} \quad \frac{1}{r} = \frac{\cos \theta}{s}$$

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2} = \frac{\mu_0 I}{4\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^2 \theta}{s^2} \frac{s}{\cos^2 \theta} \cos \theta d\theta \\ &= \frac{\mu_0 I}{4\pi s} \sin \theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\mu_0 I}{2\pi s} (= 2 \times 10^{-7} \frac{I}{s} \text{ Tesla}) \end{aligned}$$

The Equations of Magnetostatics and Ampère's Law

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{x}') \times \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} d^3x'.$$

can be expressed differently, since

$$\mathbf{B}(\mathbf{x}) = -\frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{x}') \times \nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x',$$

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \nabla \times \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x',$$

if we set $\psi = 1/|\mathbf{x} - \mathbf{x}'|$ and $\mathbf{a} = \mathbf{J}(\mathbf{x}')$, with $\nabla \times \mathbf{J}(\mathbf{x}') = 0$, since the current density is independent of $\mathbf{x}$. It follows from equation that the magnetic induction is divergence-less

$$\boxed{\nabla \cdot \mathbf{B} = 0}$$

Taking the curl of the $\mathbf{B}$ field, remembering that $\nabla \times (\nabla \times \mathbf{a}) = \nabla(\nabla \cdot \mathbf{a}) - \nabla^2 \mathbf{a}$, we get

$$\begin{aligned}\nabla \times \mathbf{B} &= \frac{\mu_0}{4\pi} \nabla \times \left(\nabla \times \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \right) \\ &= \frac{\mu_0}{4\pi} \left[\nabla \int \mathbf{J}(\mathbf{x}') \cdot \nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x' - \int \mathbf{J}(\mathbf{x}') \nabla^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x' \right]\end{aligned}$$

and using equation

$$\nabla \times \mathbf{B} = -\frac{\mu_0}{4\pi} \nabla \int \mathbf{J}(\mathbf{x}') \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x' + \mu_0 \mathbf{J}(\mathbf{x}).$$

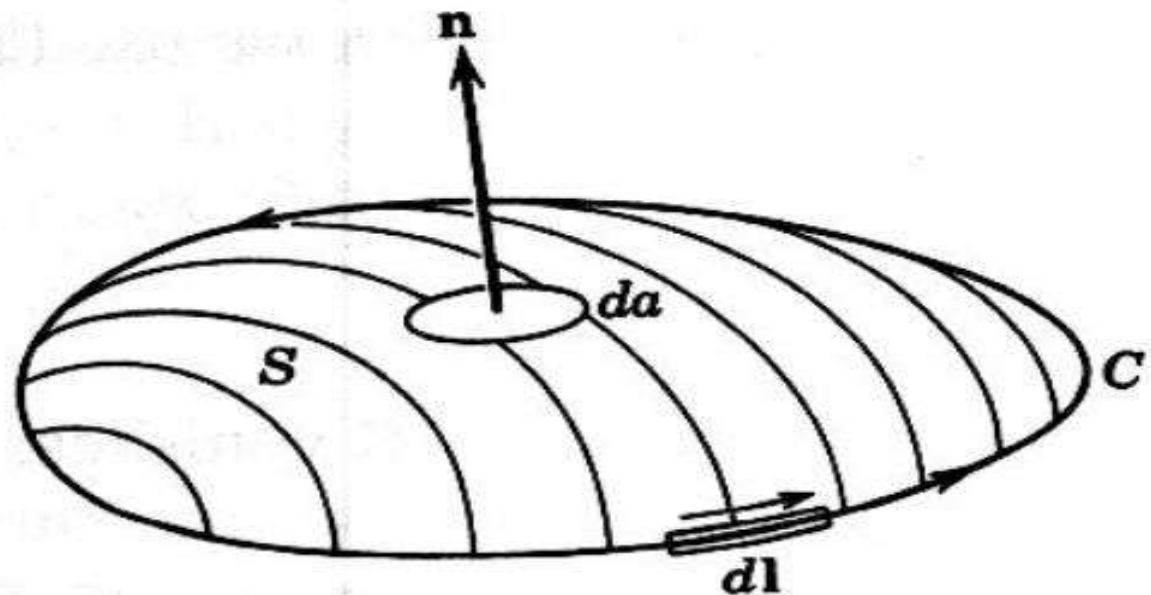
This equation can be further simplified by using the divergence theorem

$$\begin{aligned}\int \mathbf{J}(\mathbf{x}') \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x' &= \int \nabla' \cdot \left(\frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right) d^3x' - \int \frac{\nabla' \cdot \mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ &= \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \cdot \mathbf{n} da' \\ &= 0,\end{aligned}$$

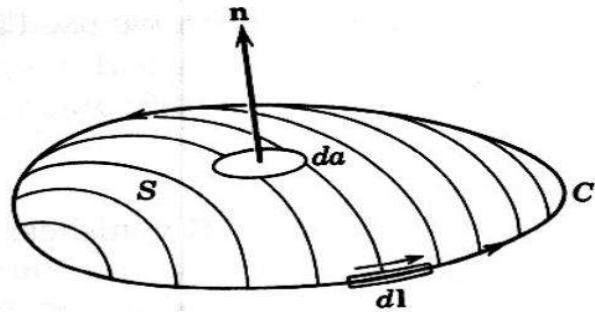
since $\nabla' \cdot \mathbf{J}(\mathbf{x}') = 0$ and the surface of integration extends over all space where the integrand vanishes. We therefore find that

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

A magnetic induction is induced around a circuit C by a current density J through the open surface S bounded by C .



If we evaluate the integral on both sides of equation over an open surface S of contour C (as Figure), we find



$$\int_S (\nabla \times \mathbf{B}) \cdot \mathbf{n} \, da = \mu_0 \int_S \mathbf{J} \cdot \mathbf{n} \, da,$$

and using Stokes' theorem

$$\boxed{\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I}$$

where I is the current passing through the surface. Equations mathematical representations of **Ampère's Law**.

Magnetic Monopoles

$\mathbf{B} = \nabla \times \mathbf{A}$. But we never paused to think about what this equation is actually telling us. In fact, it has a very simple interpretation: it says that there are no magnetic charges. A point-like magnetic charge g would source the magnetic field, giving rise a $1/r^2$ fall-off

$$\mathbf{B} = \frac{g\hat{\mathbf{r}}}{4\pi r^2}$$

An object with this behaviour is usually called a magnetic monopole.

The Vector Potential

The divergence of the magnetic induction is zero, it follows that the $\mathbf{B}$ field can be expressed as the curl of another vector field. Introducing the potential vector $\mathbf{A}(\mathbf{x})$, we can write

$$\boxed{\mathbf{B}(\mathbf{x}) = \nabla \times \mathbf{A}(\mathbf{x})}$$

Referring to equation (3.16), we find that the most general equation for $\mathbf{A}$ is

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' + \nabla \psi(\mathbf{x}),$$

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}}$$

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J},$$

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$$

The Vector Potential

We defined:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

There are infinitely many $\mathbf{A}$ that give the same $\mathbf{B}$.
This freedom is called Gauge Freedom

But here's the key issue:

☞ $\mathbf{A}$ is NOT unique

Because if we add any gradient:

$$\mathbf{A}' = \mathbf{A} + \nabla\psi$$

Then:

$$\nabla \times \mathbf{A}' = \nabla \times \mathbf{A}$$

(since curl of gradient = 0)

Coulomb Gauge (Simplification)

We choose:

$$\nabla \cdot \mathbf{A} = 0$$

- Divergence measures: “How much a field spreads out from a point”
- $\mathbf{A}$ has no sources or sinks and It behaves like an incompressible flow

Quantity	Electrostatics	Magnetostatics
Source	Charge (ρ)	Current ($\mathbf{J}$)
Potential	Scalar (V)	Vector ($\mathbf{A}$)
Equation	$\nabla^2 V = -\rho/\epsilon_0$	$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$$

$$\nabla \cdot \mathbf{A} = 0$$

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

**This is a Poisson equation
(same form as electrostatics)**

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

INDUCTANCE AND MUTUAL INDUCTANCE

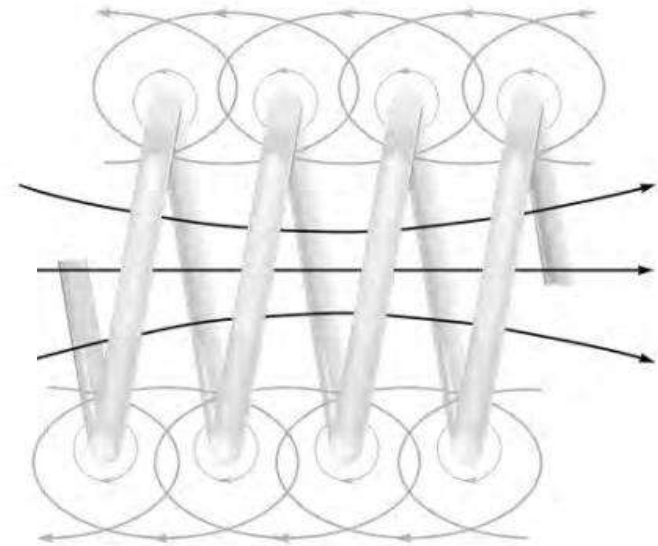
we first need to introduce the concept of flux linkage. Consider a toroid of N turns in which a current I produces a total flux Φ . We assume first that this flux links or encircles each of the N turns, and we also see that each of the N turns links the total flux Φ . The flux linkage $N\Phi$ is defined as the product of the number of turns N and the flux Φ linking each of them. For a coil having a single turn, the flux linkage is equal to the total flux.

We now define *inductance* (or self-inductance) as the ratio of the total flux linkages to the current which they link,

$$L = \frac{N\Phi}{I}$$

The current I flowing in the N -turn coil produces the total flux Φ and $N\Phi$ flux linkages, where we assume for the moment that the flux Φ links each turn. This definition is applicable only to magnetic media which are linear, so that the flux is proportional to the current. If ferromagnetic materials are present, there is no single definition of inductance which is useful in all cases, and we shall restrict our attention to linear materials.

A portion of a coil showing partial flux linkages. The total flux linkages are obtained by adding the fluxes linking each turn.



flux at the mean radius times the total number of turns. In order to obtain the total flux linkages we must look at the coil on a turn-by-turn basis.

$$\begin{aligned}(N\Phi)_{\text{total}} &= \Phi_1 + \Phi_2 + \cdots + \Phi_i + \cdots + \Phi_N \\ &= \sum_{i=1}^N \Phi_i\end{aligned}$$

where Φ_i is the flux linking the i th turn. Rather than doing this, we usually rely on experience and empirical quantities called winding factors and pitch factors to adjust the basic formula to apply to the real physical world.

Vector Potential and Inductance

An equivalent definition for inductance may be made using an energy point of view,

$$L = \frac{2W_H}{I^2}$$

where I is the total current flowing in the closed path and W_H is the energy in the magnetic field produced by the current.

express the potential energy W_H in terms of the magnetic fields,

$$L = \frac{\int_{\text{vol}} \mathbf{B} \cdot \mathbf{H} \, dv}{I^2}$$

and then replace $\mathbf{B}$ with $\nabla \times \mathbf{A}$,

$$L = \frac{1}{I^2} \int_{\text{vol}} \mathbf{H} \cdot (\nabla \times \mathbf{A}) \, dv$$

The vector identity

$$\nabla \cdot (\mathbf{A} \times \mathbf{H}) \equiv \mathbf{H} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{H})$$

may be proved by expansion in rectangular coordinates. The inductance is then

$$L = \frac{1}{I^2} \left[\int_{\text{vol}} \nabla \cdot (\mathbf{A} \times \mathbf{H}) \, dv + \int_{\text{vol}} \mathbf{A} \cdot (\nabla \times \mathbf{H}) \, dv \right]$$

After applying the divergence theorem to the first integral and letting $\nabla \times \mathbf{H} = \mathbf{J}$ in the second integral, we have

$$L = \frac{1}{I^2} \left[\oint_S (\mathbf{A} \times \mathbf{H}) \cdot d\mathbf{S} + \int_{\text{vol}} \mathbf{A} \cdot \mathbf{J} \, dv \right]$$

The surface integral is zero, as the surface encloses the volume containing all the magnetic energy, and this requires that $\mathbf{A}$ and $\mathbf{H}$ be zero on the bounding surface. The inductance may therefore be written as

$$L = \frac{1}{I^2} \int_{\text{vol}} \mathbf{A} \cdot \mathbf{J} \, dv$$

Equation expresses the inductance in terms of an integral of the values of $\mathbf{A}$ and $\mathbf{J}$ at every point. Because current density exists only within the conductor, the integrand is zero at all points *outside* the conductor, and the vector magnetic potential need not be determined there. The vector potential is that which arises from the current $\mathbf{J}$, and any other current source contributing a vector potential field in the region of the original current density is to be ignored for the present. Later we will see that this leads to a *mutual inductance*.

The vector magnetic potential $\mathbf{A}$ due to $\mathbf{J}$ is given by Eq. (51), Chapter 7,

$$\mathbf{A} = \int_{\text{vol}} \frac{\mu \mathbf{J}}{4\pi R} dv$$

and the inductance may therefore be expressed more basically as a rather formidable double volume integral,

$$L = \frac{1}{I^2} \int_{\text{vol}} \left(\int_{\text{vol}} \frac{\mu \mathbf{J}}{4\pi R} dv \right) \cdot \mathbf{J} dv$$

A slightly simpler integral expression is obtained by restricting our attention to current filaments of small cross section for which $\mathbf{J} dv$ may be replaced by $I d\mathbf{L}$ and the volume integral by a closed line integral along the axis of the filament,

$$\begin{aligned} L &= \frac{1}{I^2} \oint \left(\oint \frac{\mu I d\mathbf{L}}{4\pi R} \right) \cdot I d\mathbf{L} \\ &= \frac{\mu}{4\pi} \oint \left(\oint \frac{d\mathbf{L}}{R} \right) \cdot d\mathbf{L} \end{aligned}$$

$$L = \frac{1}{I} \oint \mathbf{A} \cdot d\mathbf{L}$$

For a small cross section, $d\mathbf{L}$ may be taken along the center of the filament. We now apply Stokes' theorem and obtain

$$L = \frac{1}{I} \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

or

$$L = \frac{1}{I} \int_S \mathbf{B} \cdot d\mathbf{S}$$

or

$$L = \frac{\Phi}{I}$$

Retracing the steps by which $L = \frac{\Phi}{I}$ is obtained, we should see that the flux Φ is that portion of the total flux that passes through any and every open surface whose perimeter is the filamentary current path.

If we now let the filament make N identical turns about the total flux, an idealization that may be closely realized in some types of inductors, the closed line integral must consist of N laps about this common path

$$L = \frac{N\Phi}{I}$$

The flux Φ is now the flux crossing any surface whose perimeter is the path occupied by any *one* of the N turns. The inductance of an N -turn coil may still be obtained

Mutual Inductance

We conclude by defining the *mutual inductance* between circuits 1 and 2, M_{12} , in terms of mutual flux linkages,

$$M_{12} = \frac{N_2 \Phi_{12}}{I_1}$$

where Φ_{12} signifies the flux produced by I_1 which links the path of the filamentary current I_2 , and N_2 is the number of turns in circuit 2. The mutual inductance, therefore, depends on the magnetic interaction between two currents. With either current alone, the total energy stored in the magnetic field can be found in terms of a single inductance, or self-inductance; with both currents having nonzero values, the total energy is a function of the two self-inductances and the mutual inductance. In terms of a mutual energy, it can be shown that (63) is equivalent to

$$M_{12} = \frac{1}{I_1 I_2} \int_{\text{vol}} (\mathbf{B}_1 \cdot \mathbf{H}_2) dv$$

or

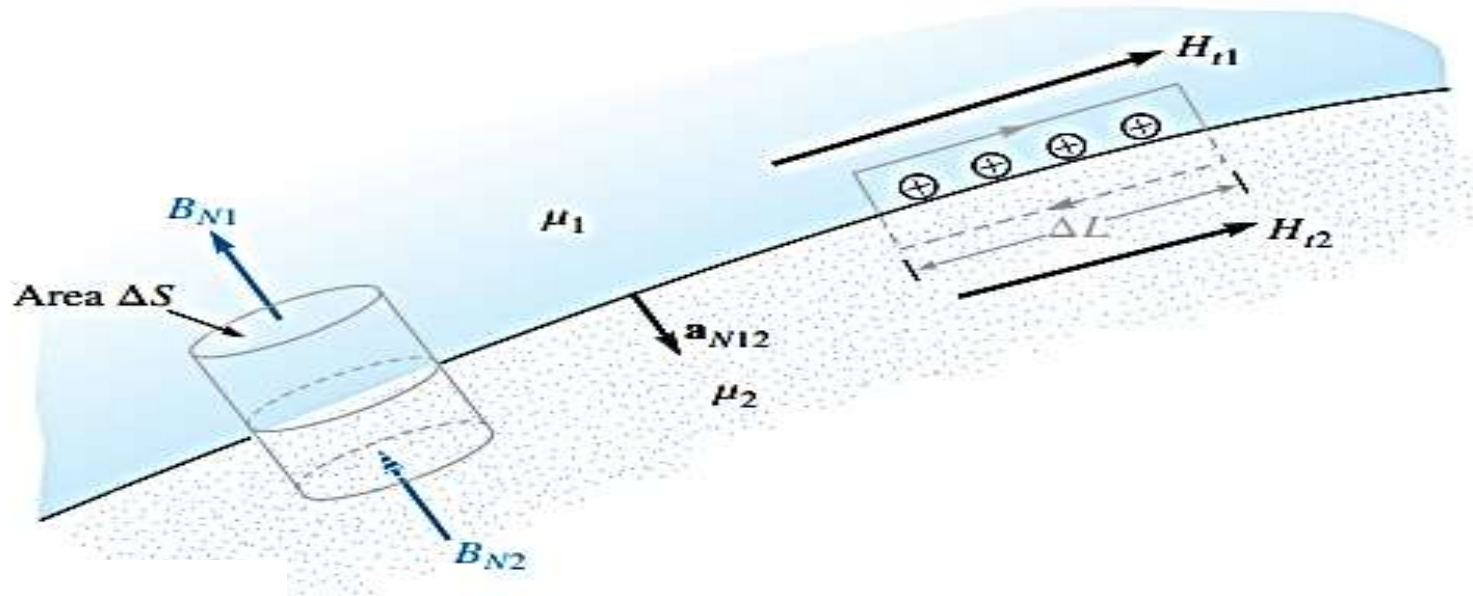
$$M_{12} = \frac{1}{I_1 I_2} \int_{\text{vol}} (\mu \mathbf{H}_1 \cdot \mathbf{H}_2) dv$$

where $\mathbf{B}_1$ is the field resulting from I_1 (with $I_2 = 0$) and $\mathbf{H}_2$ is the field arising from I_2 (with $I_1 = 0$). Interchange of the subscripts does not change the right-hand side of

$$M_{12} = M_{21}$$

MAGNETIC BOUNDARY CONDITIONS

A boundary between two isotropic homogeneous linear materials with permeabilities μ_1 and μ_2 . The boundary condition on the normal components is determined by allowing the surface to cut a small cylindrical Gaussian surface.



A gaussian surface and a closed path are constructed at the boundary between media 1 and 2, having permeabilities of μ_1 and μ_2 , respectively. From this we determine the boundary conditions $B_{N1} = B_{N2}$ and $H_{t1} - H_{t2} = K$, the component of the surface current density directed into the page.

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

we find that

$$B_{N1} \Delta S - B_{N2} \Delta S = 0$$

or the normal component of $\mathbf{B}$ is continuous across the boundary ($B_{N2} = B_{N1}$). A more formal statement of this is

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{a}_{N12} = 0$$

where $\mathbf{a}_{N12}$ is the unit normal at the boundary directed from region 1 to region 2.

The normal component of $\mathbf{B}$ is continuous, and therefore the normal component of $\mathbf{H}$ is discontinuous by the ratio μ_1/μ_2 .

$$H_{N2} = \frac{\mu_1}{\mu_2} H_{N1}$$

The relationship between the normal components of $\mathbf{M}$, of course, is fixed once the relationship between the normal components of $\mathbf{H}$ is known. For linear magnetic materials, the result is written simply as

$$M_{N2} = \chi_{m2} \frac{\mu_1}{\mu_2} H_{N1} = \frac{\chi_{m2} \mu_1}{\chi_{m1} \mu_2} M_{N1}$$

Next, Ampère's circuital law

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

is applied about a small closed path in a plane normal to the boundary surface, as shown to the right in Figure . Taking a clockwise trip around the path, we find that

$$H_{t1} \Delta L - H_{t2} \Delta L = K \Delta L$$

where we assume that the boundary may carry a surface current $\mathbf{K}$ whose component normal to the plane of the closed path is K . Thus

$$H_{t1} - H_{t2} = K$$

The directions are specified more exactly by using the cross product to identify the tangential components,

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{N12} = \mathbf{K}$$

An equivalent formulation in terms of the vector tangential components may be more convenient for $\mathbf{H}$:

$$\mathbf{H}_{t1} - \mathbf{H}_{t2} = \mathbf{a}_{N12} \times \mathbf{K}$$

Analogy with Electric circuits

Similarities:			
Electric circuit		Magnetic circuit	
Quantity	Unit	Quantity	Unit
EMF ($E=IR$)	Volt (V)	MMF ($F=\phi S$)	Ampere-turns
Current (I)	Ampere (A)	Flux (ϕ)	Weber (Wb)
Current density (J)	A/ m ²	Flux density (B)	Wb / m ² or Tesla
Resistance (R)	Ohm (Ω)	Reluctance (S)	Ampere-turns/Wb
Electric field strength (E)	Volts/m	Magnetic field strength (H)	Ampere-turns/m
Conductivity (σ) $\sigma=1/RA$	Siemen/m	Permeability, μ $\mu=1/SA$	Henry/m

'l' is the length and 'A' is the area of cross section of the conductor

Sr. No.	Magnetic Circuit	Electric Circuit
1	The closed path traced by magnetic flux is called as Magnetic circuit	The closed path traced by electric current is called as Electric circuit
2	Ohm's law – flux is the ratio of mmf to reluctance	Ohm's law – current is the ratio of emf to resistance
3	Flux is lines of forces & passes from N-pole to S-pole in magnetic circuit in Wb	Current is flow of electrons from positive to negative in electric circuit in Amp
4	MMF is driving force in AmpTurns	EMF is driving force in Volts
5	Reluctance opposes flow of flux in AT / Wb	Resistance opposes flow of current in ohms
6	Permeance is reciprocal of reluctance in Wb / AT	Conductance is reciprocal of resistance in mho
7	Flux density is the ratio of flux per cross sectional area in Wb / m ²	Current density is the ratio of current per cross sectional area in Amp / m ²
8	Reluctance is directly proportional to length of magnetic circuit	Resistance is directly proportional to length of conductor
9	Reluctivity = $\frac{1}{\mu}$ in Meter / Henry	Resistivity = ρ in ohm meter
10	Permeability = μ Henry / meter	Conductivity = σ in mho meter
11	Permanent magnet gives constant mmf	DC battery gives constant emf
12	Magnetic field strength is the ratio of mmf per length of magnetic circuit	Electric field strength is the ratio of emf per length of conductor

DISSIMILARITY

Sr. No.	Magnetic Circuit	Electric Circuit
1	Magnetic flux does not flow in circuit	Electric current actually flows in electric circuit
2	There is no magnetic insulator	There are number of electric insulators
3	Magnetic lines of forces are closed lines	Electric lines of forces are not closed lines
4	Magnetic circuit is always a closed circuit	Electric circuit can be closed or open circuit
5	Magnetic energy is initially needed to create the magnetic flux but not required to maintain it	Electric energy is continuously required to maintain the current
6	Flux can pass through air also	Current can not pass through air
7	There is a special quantity $\mu_0 = 4 \times \pi \times 10^{-7} \text{ H / m}$	There is no quantity analogous to μ_0

Magneto Motive Force, MMF (F)

- Force which drives the magnetic lines of force through a magnetic circuit
- $MMF, F = \Phi S$, where ' Φ ' is the magnetic flux and ' S ' is the Reluctance of the magnetic path.
Analogy: EMF, $V=IR$
- Also, For Electromagnets:
 $MMF = NI$ (No. of turns * Current),
where N is the number of turns of the coil and I is the current flowing in the coil
- Unit: AT (Ampere Turns)

Reluctance [S]

- Opposition of a magnetic circuit to the setting up of magnetic flux in it.

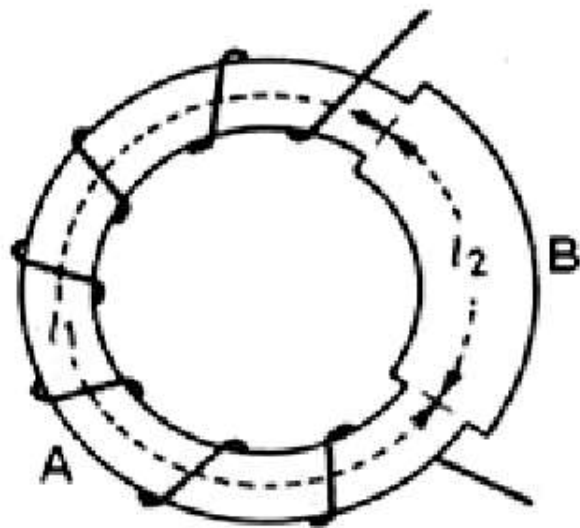
$$\text{Flux} = \phi = BA; \quad F = \text{mmf} = Hl; \quad B = \mu H$$

$$\frac{\phi}{F} = \frac{BA}{Hl} = \frac{\mu_0 \mu_r A}{l}; \quad \text{Hence} \quad \phi = \left(\frac{\mu_0 \mu_r A}{l} \right) F$$

$$\phi = \frac{F}{\left(\frac{l}{\mu_0 \mu_r A} \right)} = \frac{F}{S}; \quad \text{where } S = \left(\frac{l}{\mu_0 \mu_r A} \right)$$

- $S = F/\phi$
- Unit: AT / Wb

Analogy: Resistance



$$\mathcal{R}_1 = \frac{l_1}{\mu_1 A_1}$$

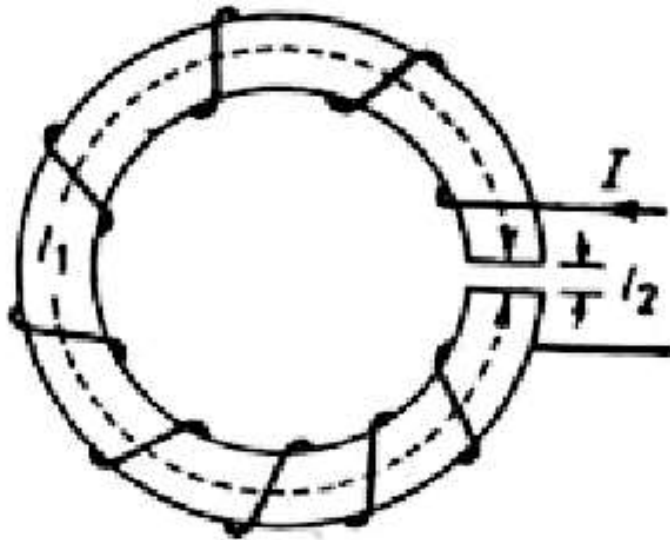
$$\mathcal{R}_2 = \frac{l_2}{\mu_2 A_2}$$

$$\therefore \text{Total Reluctance, } \mathcal{R} = \mathcal{R}_1 + \mathcal{R}_2 = \frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2}$$

$$\therefore \text{Total flux, } \Phi = \frac{\text{mmf of coil}}{\text{total reluctance}}$$

$$= \frac{F}{\mathcal{R}} = \frac{NI}{\frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2}}$$

Case 2 : (with air gap)



Total reluctance,

$$\begin{aligned}\mathcal{R} &= \frac{l_1}{\mu_1 A} + \frac{l_2}{\mu_0 A} \\ &= \frac{1}{\mu_0 A} \left(\frac{l_1}{(\mu_1 / \mu_0)} + l_2 \right) \\ &= \frac{1}{\mu_0 A} \left(\frac{l_1}{\mu_r} + l_2 \right)\end{aligned}$$

Steps to solve a problem on magnetic circuit

